

C1 implements I1

Program Code:

```
package core_java;

// Interface I1
interface I1 {
    void display();

    default void startSystem() {
        System.out.println("Initializing system hardware...");
        logAction();
    }

    // Private method inside the interface
    private void logAction() {
        System.out.println("Internal Log: Connection verified successfully.");
    }
}

// Class C1 implementing Interface I1
class C1 implements I1 {
    // Overriding the abstract method
    public void display() {
        System.out.println("Class C1 dashboard is now active.");
    }
}

// Execution Class
public class C1toI1 {
    public static void main(String[] args) {
        C1 obj = new C1();

        obj.startSystem();
        obj.display();
    }
}
```

Explanation:

This program demonstrates interface implementation. Interface I1 contains the abstract method `display()`, a default method `startSystem()`, and a private method `logAction()`. Class C1 implements I1 and provides the implementation of `display()`. The `startSystem()` method calls the private `logAction()` method internally, and the C1 object calls both `startSystem()` and `display()`.

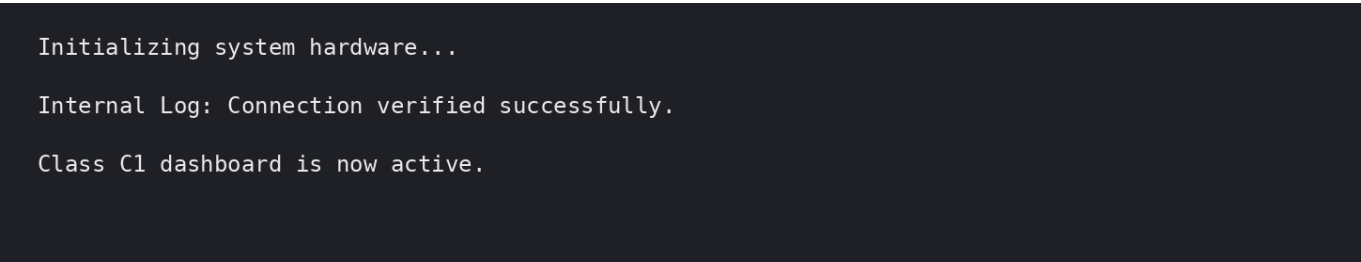
Compilation and Execution Commands:

Save the file as: C1toI1.java

Compile:
`javac -d . C1toI1.java`

Run:
`java core_java.C1toI1`

Output Screenshot:



```
Initializing system hardware...

Internal Log: Connection verified successfully.

Class C1 dashboard is now active.
```

Verification:

The output confirms that the default method `startSystem()` executes successfully, calls the private interface method internally, and the C1 implementation of `display()` is also executed.

Result:

Thus, the Java program was successfully executed and the relationship C1 implements I1 was verified successfully.